Build a lightweight inference logging and ingestion system for an LLM application.

1. Chatbot Application

Build a simple chatbot using any foundation model API.

Examples:

- GPT-4.1
- Claude Sonnet
- Gemini
- DeepSeek
- Grok
- any equivalent model

The chatbot should:

- support multi-turn conversations
- maintain short conversational context
- expose a simple UI

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2. Lightweight SDK / Wrapper

Create a lightweight SDK, middleware, or wrapper around your LLM calls that captures inference metadata.

Examples of metadata:

- model
- provider
- latency
- token usage
- timestamps
- request status/errors
- conversation/session ID
- input/output previews

The SDK should send logs to an ingestion endpoint in near real time. Implementation details are flexible.

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3. Ingestion Pipeline

Build an ingestion service/API that:

- receives logs from the SDK
- validates/parses payloads
- extracts useful metadata
- stores processed data in a database

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4. Database Storage

Store:

- chat messages
- inference logs
- extracted metadata

We care about sensible schema design and practical tradeoffs.

Deliverables

1. GitHub Repository

Complete source code.

2. README

Include:

- setup instructions
- architecture overview
- schema design decisions
- tradeoffs made
- what you would improve with more time

3. Architecture Notes

Briefly explain:

- ingestion flow
- logging strategy
- scaling considerations
- failure handling assumptions

4. Demo

Hosted link, screenshots, or Loom video.

Bonus

You will be given a guaranteed interview if you are able to complete the following task.

- **Multi-provider support**
- **Streaming Responses**
- **Latency + Throughput + Errors dashboards**
- **Docker Compose one-command setup**
- **Event based architecture**
- PII redaction
- Deploy application on self hosted k8s

Frontend

The UI allows following:

1. Cancel a conversation
2. List conversations
3. Resume a conversation

Submission

Please send:

- GitHub repo
- architecture notes
- demo link (optional)

to: **work@ollive.ai**

Looking forward to seeing what you build 🚀